END-TO-END DATA ANALYTICS PROJECT

Kister Retail Sales – Business Analysis

Excel • Power Query • SQL • ML • Power BI

An end-to-end retail analytics project that examines sales and profit by region, product category, product and city, and looks ahead with a six-month sales forecast. The work follows the full analytics lifecycle: data understanding, cleaning in Excel and Power Query, SQL investigation, machine-learning forecasting, KPI definition, and an interactive Power BI report.



Power BI report visual used as the presentation layer of the broader analysis.

PROJECT SUMMARY

Business domain	Retail sales and profit performance	**Analysis focus**	Region, category, product, city, time
Data tools	Excel, Power Query and SQL for preparation and validation; ML for forecasting	**Reporting tool**	Power BI with DAX measures and slicers
Intended users	Business owners, sales managers, regional teams	**Outcome**	Evidence-led comparisons and performance monitoring

Project goal: move from raw retail records to validated analysis and a clear business-facing report.

01 | Project Overview

Kister Retail Sales – Business Analysis studies sales through region, category, product and city dimensions. The purpose is not only to show charts, but to follow a workflow that checks data quality, explores patterns, validates calculations, and turns findings into information business teams can act on.

Business problem

Sales and profit data is hard to interpret when it sits in raw files. Managers need one place to see how much the business sells, how profitable it is, which regions, categories, products and cities drive results, and what sales may look like next.

Key analytical questions

- What are the total sales, total profit, profit margin and number of orders?
- Which regions contribute the largest and smallest share of sales?
- Which product categories earn the most sales and profit?
- Which products and cities lead on sales?
- How do results change by year and month?
- What are the expected sales for the next 6 months?
- Do results stay consistent across Excel, SQL and Power BI?

Scope of work

STAGE	ACTIVITIES	OUTPUT
Data understanding	Inspect columns, data types, row grain and category definitions.	Data dictionary and quality notes
Data preparation	Excel and Power Query: fix types, remove duplicates, handle blanks, standardise labels.	Clean, consistent dataset
SQL investigation	Filter, group and aggregate sales and profit by region, category, product and city.	Reusable queries and summaries
ML forecasting	Model monthly sales trend to project the next 6 months.	Forecast of future sales
BI reporting	Build DAX measures, slicers and visuals in Power BI.	Interactive 8-page report
Validation and communication	Cross-check totals across tools and write findings.	Validated metrics and summary

02 | Tools & Analytical Methods

TOOL	PURPOSE IN THE PROJECT
Excel	First look at the data, quick pivots, manual checks and data-quality review.
Power Query	Repeatable cleaning: data types, null handling, column cleanup and category standardisation.
SQL	Filtering, GROUP BY and aggregate queries to validate totals by region, category, product and city.
ML	Time-series forecasting of monthly sales for the next 6 months.
Power BI + DAX	KPI cards, bar and donut charts, forecast line, slicers; measures for Total Sales, Total Profit, Profit Margin % and Orders.

Analysis workflow

1. Understand the dataset

Document what one row represents and which fields are used for grouping, filtering and calculation.

2. Prepare the data

Clean and standardise in Excel and Power Query; check duplicates, blanks and ranges.

3. Investigate in SQL

Build grouped summaries by region, category, product, city, year and month; compare with Excel results.

4. Forecast with ML

Fit a trend model on monthly sales and project the next 6 months.

5. Build the Power BI model

Load clean data and use DAX measures instead of hard-coded KPI values.

6. Visualise and test

Check slicers, cross-filtering and totals against the validated analysis.

Expected deliverables

- A cleaned and documented dataset.
- Excel and Power Query preparation steps.
- SQL queries for KPI validation and segment analysis.
- ML forecast of future sales.
- Power BI report with measures, filters and visual comparisons.

03 | Metrics, Interpretation & Portfolio Summary

Metrics shown in the report

TOTAL SALES	TOTAL PROFIT	PROFIT MARGIN %	ORDERS
3.43M	1.09M	0.32	1,223

What the dashboard shows

- **West** leads sales at 1.09M (31.69%), then South 0.99M (28.77%) and North 0.85M (24.83%). **East** is lowest at 0.50M (14.71%).
- Electronics and Clothing are the largest categories by sales; Stationery is the smallest.
- Top products: Bluetooth H… and Sports Shoes at 0.29M each, then Power Bank 0.25M and Office Chair 0.24M.
- Top city: Ahmedabad at 0.32M, then Chandigarh and Bengaluru at 0.27M each.
- The forecast visual projects sales for the next 6 months to support planning.

How to interpret the analysis responsibly

- Use region and category comparisons to find patterns, not to claim causes without more evidence.
- Profit Margin % of 0.32 means profit is about 32% of sales.
- Confirm whether Orders counts order IDs or transaction rows.
- Treat the forecast as an estimate that depends on past trends.

Portfolio website description

Kister Retail Sales – Business Analysis. An end-to-end retail analytics project using Excel, Power Query, SQL, machine learning and Power BI to examine sales and profit across regions, product categories, products and cities. The workflow covers data cleaning, SQL-based aggregation and validation, KPI definition, a six-month sales forecast, and an interactive Power BI report that helps stakeholders explore performance and spot areas for further analysis.

> **Before sharing:** list only the tools you actually used; confirm currency, field names and aggregation rules; and re-check the figures against the source data.